Einstein's brainchild,[fn2] and it fit nicely into his equations of gravity governing the evolution of the universe. But it was based on a deeply mistaken notion, and by rights it should never have been written down in the first place.

Einstein's heart was in the right place. The purpose of the cosmological constant was to save the universe from catastrophic collapse. Or more accurately, from having catastrophically collapsed already. Being an expert, as he was, in all things gravitational, Einstein knew that all the data available pointed to the uncomfortable conclusion that gravity should have destroyed the universe long ago. This was 1917, half a century before widespread acceptance of the Big Bang theory, when the cosmos was still largely thought to be static and unchanging. Stars could live and die, matter might slightly rearrange, but space was *space*—it was just a background on which other things happened. So when Einstein saw that there were stars in the night sky, apparently stationary, he knew the universe was in trouble. Every one of those stars, he figured, should be gravitationally attracting every other star, and slowly drawing together over time. It doesn't help for the other stars to be really distant, either; gravity is an infinite and purely attractive force. (It should be noted that this was a time before it was clear that other galaxies existed, or he would have applied the argument to galaxies instead. The problem would be the same.) In an unchanging universe, you can never be far enough away from something to not feel its pull, at some level, and that pull should be bringing you together over time. Einstein's own calculations said that any universe populated with massive objects should already have collapsed upon itself. The very existence of the cosmos was a contradiction.

This, of course, looked bad. Fortunately, Einstein found room in his General Theory of Relativity to add a little universe-rescuing tweak. Nothing in space could counter the gravity of the stars, but perhaps *space itself* could do it. Einstein had already developed a beautiful equation to describe how the shape of space responded to the gravitational attraction of all the stuff in the cosmos. All he had to do to ensure gravitational attraction wouldn't immediately collapse space was to decide his equation was incomplete, and to tack on a term that could stretch out the space between gravitating objects, perfectly balancing the contraction that gravity would otherwise cause. The term didn't represent a new component of the universe, but a property of space itself, where each piece of space has a